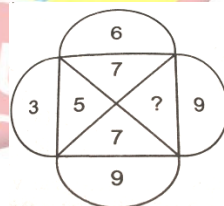




- Bala had three sons. He had some chocolates which he distributed among them. To his eldest son, he gave 3 chocolates more than half the number of chocolates with him. To his second eldest son, he gave 4 chocolates more than one-third of the remaining number of chocolates more with him. To his youngest son, he gave 4 chocolates more than one-fourth of the remaining number of chocolates with him. He was left with 11 chocolates. How many chocolates did he initially have ?
(a)180 (b)78 (c)144 (d)120
- How long would it take you to count 1 billion orally if you could count 200 every minute and were given a day off every four years? Assume that you start counting on 1 January 2001.
(a) 10 yr, 107 days, 5 h, 20 min
(b) 8yr, 287 days, 15h, 20 min
(c) 9 yr, 187 days, 5 h, 20 min
(d) 9 yr, 278 days, 12 h, 34 min
- Find the value of 'x', if $\left(\frac{1}{2^{\log_x 4}}\right)\left(\frac{1}{2^{\log_x 16}}\right)\left(\frac{1}{2^{\log_x 256}}\right) \dots \infty = 2$
(a)2 (b)1/2 (c)4 (d)1/4
- Find the unit digit of $(13687)^{3265}$
(a)1 (b)3 (c)7 (d)9
- How many pairs of letters are there in the word 'PRISON', each of which has as many letters between its two letters in the word as there are between them in the English alphabet?
(a)Two (b)One (c)Four (d)Three
- Twelve villages in a district are divided into 3 zones with 4 villages per each zone. The telephones department of the district intends to connect the villages with telephone lines such that every two villages in the same zone are connected with three direct lines and every two villages belonging to different zones are connected with two direct lines. How many direct lines are required?
(a)210 (b)96 (c)54 (d)150
- Cars are safer than planes. Fifty per cent of plane accidents results in death, while only one percent of car accidents result in death. Which of the following, if true, would most seriously weaken the argument above?
(a) Planes are inspected more often than cars.
(b) The number of car accidents is several hundred thousand times higher than the number of plane accidents.
(c) Pilots never fly under the influence of alcohol, while car drivers often do.
(d) Plane accidents are usually the fault of air traffic controllers, not of pilots.
- Sum of all three digit numbers (no digit being zero) having the property that all digits are perfect squares is
(a)3108 (b)6216
(c)13986 (d)None of these
- A teacher gave a student the task of adding 'N' natural numbers starting from 1. After a while, the student reported his result as 700. The teacher replied that his result was wrong. The student realized that he has added one number 24r twice by mistake. Find the sum of the digits of the number which the student had added twice.
(a)5 (b)6 (c)7 (d)8
- Computer A takes 3 min to process an input while computer B takes 5 min. If computers A, B and C can process an average of 14 inputs in one h, how many minutes does computer C alone take to process one input?
(a)10 (b)4 (c)6 (d)None

- Arrange the following statements P, Q, R and S in a logical order to make a sensible paragraph.
P: The logic that underlines this is two-fold.
Q: RBI is likely to cut bank rate and CRR.
R: Over the last few years, RBI has tried to implement policy changes.
S: According to sources, bank rate will be cut in April.
(a)QRP S (b)RPSQ (c)QSRP (d)SPRQ
- If $A + B = C + D$, $B + D = 24$, $D + E > A + B$, $C + D > A + E$, then which of the following is true?
(a) $D > B > E > A > C$ (b) $A > B > D > E > C$
(c) $A > D > B > E > C$ (d) $D > A > B > E > C$
- Reena visited her High School friend, Natasha after their 25th school reunion. "what a nice pair of children you have, are they twins?" Reena asked.
"No my sister is older than I", said Natasha's son Rahul. "The square of my age plus the cube of her age is 7148". "The square of my age plus the cube of his age is 5274", said Preeti, Natasha's daughter.
How old were they?
(a)Preeti 23, Rahul 14 (b)Preeti 18, Rahul 16
(c)Preeti 21, Rahul 19 (d)Preeti 19, Rahul 17
- What will come in place of the question mark (?) in the following series?
12 22 69 272 1365 ?
(a)8196 (b)8195
(c)6830 (d)8184
- Divide Rs. 1074 (in whole Rs. having incremental amounts) into a number of bags so that I can ask for any amount between Rs. 1 and Rs. 1074 and you can give me the proper amount by selecting a certain number of these bags without opening them. What is the minimum number of bags you will require?
(a)12 (b)10 (c)9 (d)11
- Which number will be there in the place of question mark (?) in the following figure?



- (a)5 (b)6 (c)4 (d)8
- The remainder when $x = 1! + 2! + 3! + 100!$ is divided by 240 is
(a)153 (b)33 (c)73 (d)187
- Using the digits 1, 5, 2, 8 all possible four digit numbers is between
(a)10000 and 2000 (b)20000 and 50000
(c)50000 and 100000 (d)100000 and 150000
- The sum of the numbers from 1 to 100, which are not divisible by 3 and 5, is
(a)2946 (b)2732 (c)2632 (d)2317
- A motorboat going downstream a river passes a raft (freely floating with the river current) at a certain point 'A'. Exactly one hour later it turned back and after some time while coming upstream it passed the same raft at a point 'B' situated at a distance of 6 km from point 'A'. What is the speed of the water current in the river expressed in km/h ?
(a)2.0 (b)3.0 (c)4.0 (d)6.0
- You are in the land logic, where there are 3 types of rabbits. Blue rabbits always tell the truth, green rabbits sometimes tell the truth and red rabbits never tell the truth. Assume you cannot distinguish colours. A





rabbit says to you "I always lie". What colour of rabbit is speaking to you?

- (a) Blue (b) Red
(c) Green (d) Cannot be concluded

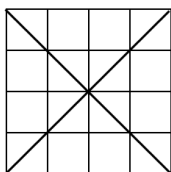
22. It is wrong for doctors to lie about their patients illnesses? Aren't doctors just like any other people we hire to do a job for us? Surely, we would not tolerate not being told the truth about the condition of our automobile from the mechanic we hired to fix it, or the condition of our roof from the carpenter we employed to repair it. Just as these workers would be guilty of violating their good faith contracts with us if they were to do this, doctors who lie to their patients about their illnesses violate these contracts as well, and this is clearly wrong. The conclusion of the argument is best expressed which of the following?

- (a) Doctor who lie to their patients about their illnesses violate their good faith contracts with their patients.
(b) Doctor often lie to their patients about their patients illnesses.
(c) It is wrong for doctors to lie about their patients illnesses.
(d) Doctor, like mechanics and carpenters, enter into good faith contracts with us when we hire them.

23. All the letters of the word 'INDIA' are permuted in all possible ways and the words so formed are written as in dictionary then the 58th word in the list is

- (a) NIIDA (b) INIDA (c) NIDIAI (d) NIDAI

24. Identify the number of triangles in the figure given below :



- (a) 44 (b) 48 (c) 36 (d) 32

Directions for question 65 : Choose the ordered pair of statements (P to S) where the first statement implies the second, and two statements are logically consistent with the main statement.

25. Each time Sachin is the captain India loses

- (P) Sachin is the captain (Q) India did not win
(R) Sachin is not the captain (S) India won
(a) PS (b) SR (c) SP (d) RP

26. If all the 6's are replaced by 9's, then the algebraic sum of all the numbers from 1 to 100 (both inclusive), varies by

- (a) 330 (b) 333 (c) 219 (d) 279

27. Recently, while in Bangalore, I decided to walk down the escalator of a tube station. I did some quick calculation in my mind. I found that if I walk down twenty six steps, I require thirty seconds to reach the bottom. However, if I am able to step down thirty four stairs I would only require eighteen seconds to get to the bottom. If the time is measured from the moment the top step begins to descend to the time I step off the last step at the bottom, what is the height of the stairway in steps?

- (a) 40 (b) 46 (c) 52 (d) 58

28. When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?

- (a) 5 (b) 6 (c) 7 (d) 8

29. Pick the 1st, 2nd, 4th, 5th and 6th letters in the word REASONING, form yet another word and then write the first and last letters of the word formed.

- (a) SE (b) ES (c) NE (d) OR

30. While Hameed had his back turned, a dog ran into his butcher shop, snatched a piece of meat off the counter and ran out. Hameed was mad when he realized what had happened. He asked other shopkeepers, who had seen the dog to describe it. The shopkeepers, really did not what to help Hameed. So, of them made a statement which contained one truth and one lie.

Shopkeeper 1 said: "The dog had black hair and a long tail".

Shopkeeper 2 said: "The dog had a short tail and wore a collar".

Shopkeeper 3 said: "The dog had white hair and no collar".

Based on the above statements which of the following could be correct description.

The dog had

- (a) white hair, short tail and no collar
(b) white hair, long tail and a collar
(c) black hair, long tail, and a collar
(d) black hair, long tail and no collar

31. A train after travelling 60 km meets with an accident and then proceeds at $\frac{3}{4}$ of its former rate and arrives at the terminus 40 minutes late. Had the accident happened 25km further on, it would have arrived 10 minutes sooner. Find the speed of the train and the distance, respectively.

- (a) 160 km / hr, 150 km (b) 160 km / hr, 140 km
(c) 50 km / hr, 160 km (d) 40 km / hr, 160 km

Directions (Q. Nos. 72-75) Read the following information carefully to answer the questions.

After months of talent searching for an administrative assistant to the president of the college the field of applicants had been narrowed down to five (A, B, C, D, and E). It was announced that the finalist would be chosen after a series of all-day group personal interviews. The examining committee agreed upon the following procedure.

- (1) The interviews will be held once a week.
- (2) Three candidates will appear at an all-day interview session.
- (3) Each candidate will appear atleast once.
- (4) If it becomes necessary to call applicants for additional interviews.
- (5) Because of a detail given in the written applications, it was agreed that whenever candidate B appears, A should also be present.
- (6) Because of travel difficulties, it was agreed that C will appear for only one interview.

32. Which of the following is a possible sequence of combinations for interviews in two successive weeks?

- (a) ABC ; BDE (b) ABD ; ABE
(c) ADE ; ABC (d) BDE ; ACD

33. At the first interview, the following candidates appear:

A, B and D. Which of the following combinations can be called for the interview to be held the next week?

- (a) BCD (b) CDE
(c) ABE (d) ABC

34. Which of the following correctly state (s) the procedure followed by the search committee?

- (I) After the second interview, all applicants might have appeared atleast once.
(II) The committee interviews each applicant atleast once.





(III) If a third session is held, it is possible for all applicants to appear atleast twice.

- (a) I only (b) II only
(c) I and II only (d) III only

35. A, B and D appear at the interview and D is called for an additional interview the following week, which two candidates may be asked to appear with D?

- (i)A (ii)B (iii)C (iv)E
(a)I and II only (b)III and IV only
(c)II and III only (d)II and IV only

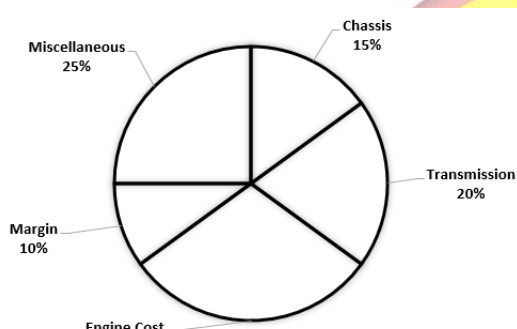
36. How many 5s are there in the following number series each of which is immediately followed by 4 but not immediately preceded by 6 ?

456 656 455 455 654 456 456 5454

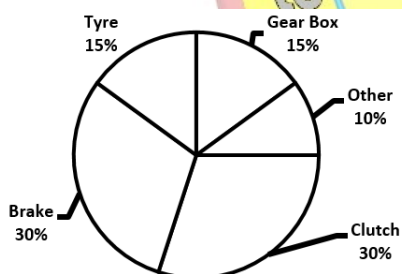
- (a)One (b)Three (c)Four (d)Two

Direction (Q. Nos. 77-81) Study the pie charts given below and answer the following questions.

SALE PRICE BREAK-UP OF CAR



COST BREAK-UP OF TRANSMISSION



Price of car = Rs. 100000

37. If transmission cost increases by 20%, by what amount is that profit reduced (total price of car remains same)?

- (a) ₹ 3000 (b) ₹ 4000
(c) ₹ 6000 (d) can't be determined

38. If the price of types goes up by 25%, by what amount should the sale price be increased to maintain the amount of profit?

- (a) ₹ 750 (b) ₹ 2250
(c) ₹ 3750 (d) ₹ 375

39. If transmission cost increase by 10% and engine cost increase by 20%, what is the percentage contribution of transmission cost with respect to the total cost?

- (a) 20% (b) 22.44% (c) 21.86% (d) 21.98%

40. If all the costs increase by 10% and the selling price remains the same, by what per cent will the profit be reduced?

- (a) 50% (b) 90%
(c) 10% (d) can't be determined

41. What is the profit percentage?

- (a) 10 (b) 9.09%
(c) 11.11% (d) can't be determined

Direction (Q. Nos. 82-84) Read the following passage to answer the questions
Rajita has a unique way of attempting the question paper having 50 questions. She starts from question 1 and attempts all questions which are in A.P. with a common difference of 3 in the forward direction and 3 in the reverse direction. If she reaches a stage when she cannot attempt any more question, she starts in the reverse direction with the first unanswered question. She repeats the same process and when she reaches a stage when she can not process any further, she reverses her direction again starting with the first unanswered question.

42. How many times does she reverse her direction?

- (a) 3 (b) 4 (c) 5 (d) 6

43. Which is the last question that she answers if she attempts all the 50 questions ?

- (a) 50 (b) 49 (c) 48 (d) 3

44. Which is the 20th question Rajita answers ?

- (a) 50 (b) 48 (c) 47 (d) 44

45. If $A_1 = \{3\}$, $A_2 = \{5, 7, 9\}$, $A_3 = \{11, 13, 15, 17, 19\}$,

$A_4 = \{21, 23, 25, 27, 29, 31, 33\}$ and so on, what is the average of the numbers of the set A_{20} ?

- (a) 761 (b) 763 (c) 765 (d) 767

Direction (Q. Nos. 86-89) Read the following passage to answer the questions

In each question below are given three statements followed by three conclusions numbered I, II and III. You have to take the three given statements to be true even if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follow(s) from the given statements disregarding commonly known facts. Then decide which of the answers (A), (B), (C) and (D) is the correct answer.

46. Statements :

Some trees are branches. All buds are branches.
All flowers are trees.

Conclusions,

- (I) Some branches are buds.
(II) Some trees are flowers..
(III) Some buds are trees.

- (a) Only I follows (b) Only II follows
(c) Only I and II follow (d) All follow

47. Statements :

Some pots are eatables.
All eatables are drinks.
No banana is pot.

Conclusions :

- (I) Some pots are drinks.
(II) All eatables are pots.
(III) Some drinks are eatables.

- (a) Only I follows (b) Only III follows
(c) Only II follows (d) Only I and III follow

48. Statements :

All jewels are rings.
Some rings are necklaces.
Some cakes are jewels.

Conclusions :

- (I) Some necklaces are jewels.
(II) Some rings are cakes.
(III) No jewel is necklace.

- (a) Only II and either I or III follow





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- (b) Only either I or III follows
 (c) Only II and III follow
 (d) Only II follows

49. Statements

All actors are writers.
 Some writers are dancers.
 All poets are writers.

Conclusions

- (I) All actors are poets
 (II) Some dancers are writers
 (III) Some dancers are actors
 (a) None follows
 (b) Only I and II follow
 (c) Only II and III follow
 (d) Only I and III follow

Direction (Q. Nos. 90-94) Read the following passage to answer the questions

Five houses lettered A, B, C, D, & E are built in a row next to each other. The houses are lined up in the order A, B, C, D, & E. Each of the five houses have coloured roofs and chimneys. The roof and chimney of each house must be painted as follows

- (1) The roof must be painted either green, red, or yellow.
 (2) The chimney must be painted either white, black or red.
 (3) No house may have the same colour chimney as the colour of roof.
 (4) No house may use any of the same colours that the every next house uses.

- (5) House E has a green roof.
 (6) House B has a red roof and a black chimney.

50. Which statement is false ?

- (a) House A has a yellow roof.
 (b) House A & C have different colour chimneys.
 (c) House D has a black chimney.
 (d) House E has a white chimney.

51. Which of the following is true ?

- (a) At least two houses have black chimneys.
 (b) At least two houses have red roofs.
 (c) At least two houses have white chimneys.
 (d) At least two houses have green roofs.

52. What is maximum total number of green roofs for houses

- (a) 1 (b) 2 (c) 3 (d) 4

53. Which possible combinations of roof & chimney can a house have ?

- (I) A red roof & a black chimney.
 (II) A yellow roof & a red chimney.
 (III) A yellow roof & a black chimney.
 (a) I & II & III (b) II only (c) III only (d) I & II only

54. If house C has a yellow roof, which one of the following is true?

- (a) House E has a white chimney.
 (b) House E has a black chimney.
 (c) House E has a red chimney.
 (d) House D has a red chimney.

55. You have 13 balls which all look identical. All the balls are of the same weight except for one. Using only a balance scale, you can find the odd one out with how many minimum number of weighing ?

- (a) 3 (b) 5 (c) 6 (d) 4

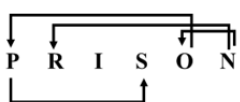
ANSWER

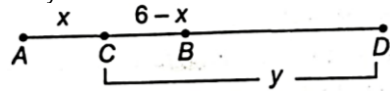
1.	B	2.	C	3.	C	4.	C	5.	C
6.	D	7.	B	8.	A	9.	C	10.	C
11.	C	12.	D	13.	D	14.	D	15.	D
16.	D	17.	A	18.	D	19.	C	20.	B
21.	C	22.	A	23.	C	24.	C	25.	B
26.	A	27.	B	28.	B	29.	C	30.	B
31.	C	32.	C	33.	B	34.	A	35.	B
36.	D	37.	B	38.	A	39.	A	40.	C
41.	B	42.	A	43.	C	44.	D	45.	B
46.	C	47.	D	48.	A	49.	A	50.	B
51.	C	52.	C	53.	A	54.	A	55.	D





SOLUTION

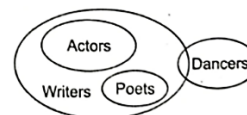
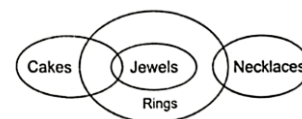
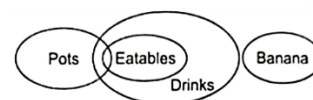
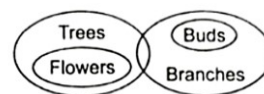
1. (b) Let x be the number of chocolates initially, Bala has.
Then, $\frac{x}{2} + 3 + \frac{1}{3}\left(\frac{x}{2} - 3\right) + 4 + \frac{1}{4}\left[\frac{x}{3} - 6\right] + 4 + 11 = x$
 $\Rightarrow \frac{3x}{4} + \frac{39}{2} = x$
 $\Rightarrow \frac{x}{4} = \frac{39}{2}$
 $\Rightarrow x = \frac{39 \times 4}{2}$
 $\Rightarrow x = 78$
2. (c) 1 billion = 10^9
Total time taken for counting = $\frac{10^9}{200} = 5 \times 10^6$ min
= 83333 h 20 min
= 3472 days 5 h 20 min
= 9 yr 187 days 5 h 20 min
3. (c) From given relation, we get
Then, $2^{\log x^4 + \log x^{16} + \log x^{256} + \dots} = \frac{1}{2}$
 $\Rightarrow 2^{\frac{1}{\log x}(\log 4 + \log 4^2 + \log 4^4 + \dots)} = \frac{1}{2}$
 $\Rightarrow 2^{\frac{\log 4}{\log x}(1 + 2 + 4 + \dots)} = \frac{1}{2}$
 $\Rightarrow 2^{\frac{\log 4}{\log x} \left(\frac{1}{1-2}\right)} = \frac{1}{2}$
 $\Rightarrow 2^{-\log x 4} = 2^{-1}$
 $\Rightarrow \log_x 4 = 1$
 $\Rightarrow x = 4$
4. (c) Power of last digit
 $7^1 = 7, 7^2 = 9$
 $7^3 = 3, 7^4 = 1$
As, $(13687)^{3265} = (13687)^{816 \times 4 + 1}$ Hence, last digit of
 $(13687)^{3265}$ is last digit of $7^1 = 7$
5. (c)
- 
- Hence, number of such pairs are PS, NO, NR, OR.
6. (d) Number of direct lines
= $({}^4C_2 \times 3) \times 3 + {}^3C_2 \times {}^4C_1 \times {}^4C_1 \times 2$
= $6 \times 3 \times 3 + 3 \times 4 \times 4 \times 2$
= $54 + 96 = 150$
7. (b) Statement b weaken the argument given.
8. (a) Digits being perfect square and non-zero include 1, 4 and 9.
At each place digit appear $2! - 2$ times, so sum of all the three digit numbers
= $(1 + 4 + 9) \times 2!(111)$
= $28(111) = 3108$
9. (c) As $1 + 2 + \dots + N = \frac{N(N+1)}{2}$
 $\frac{N(N+1)}{2}$ just less than 700 is obtained for N36 and $N(N+1)/2$ equals 666.
Thus, number $700 - 666 = 34$ was added twice. Hence, sum of digits = $3 + 4 = 7$
10. (c) Let x min be the time taken by C to process one input.
Then, $14 = \frac{20 + 12 + \frac{60}{x}}{3}$
 $\Rightarrow 42 = 32 + \frac{60}{x}$
 $\Rightarrow \frac{60}{x} = 10$
 $\Rightarrow x = 6$
11. (c) Q S R P

12. (d) $A + B = C + D$... (i)
 $B + D = 2A$... (ii)
 $D + E > A + B$... (iii)
 $C + D > A + E$... (iv)
From Eqs. (i) and (iv),
 $A + B > A + E \Rightarrow B > E$
From Eqs. (i) and (iii),
 $D + E > C + D \Rightarrow E > C$
 $\Rightarrow B > E > C$
By putting in Eq. (i), we get
 $A < D$
 $\Rightarrow D > A > B > E > C$
13. (d) Let x be the age of Rahul and y be the age of Preeti
 $\Rightarrow x^2 + y^3 = 7148$ (i)
 $x^3 + y^2 = 5248$ (ii)
On solving Eqs. (i) and (ii), we get
 $x = 17$ and $y = 19$
14. (d) $22 = 12 \times 2 - 2$
 $69 = 22 \times 3 + 3$
 $272 = 69 \times 4 - 4$
 $1365 = 272 \times 5 + 5$
 $? = 1365 \times 6 - 6 = 8184$
15. (d) Amounts required in the bag will be powers of 2 i.e., 1, 2, 4, 8, 16, 32, 64, 128, 256, and 512.
Thus, minimum number of bags required = 11
16. (d) ? will be filled by 8.
Blocks have sum,
 $3 + 9 = 12, 5 + ? = 13, 7 + 7 = 14$, and $9 + 6 = 15$
17. (a) As $6! = 720$, so numbers from 61 onwards is divisible by 240.
Hence, remainder is
 $1! + 2! + 3! + 4! + 5!$
 $= 1 + 2 + 6 + 24 + 120 = 153$
18. (d) Sum of the numbers will be
 $(1 + 5 + 2 + 8) 3! (1111) = 96(1111) = 106656$
which lies between 100000 and 150000
19. (c) Required sum
= $(1 + 2 + \dots + 100) - (3 + 6 + \dots + 99) - (5 + 10 + \dots + 100)$
+ $(15 + 30 + \dots + 90)$
= $5050 - \frac{3 \times 33 \times 34}{2} - \frac{5 \times 20 \times 21}{2} + 15 \times 21$
= $5050 - 1683 - 150 + 315 = 2632$
20. (b) Let x and y be speed of water current and motor boat respectively.
- 
- C and D are respectively the points where raft and motor boat are located when motor boat turns back.
 $\Rightarrow \frac{6-x}{x} = \frac{y+x-6}{y-x}$
 $\Rightarrow \frac{6}{x} = \frac{2y-6}{y-x}$
 $\Rightarrow x = 3$
and $y = 4$
21. (c) Only green rabbit can say this type of statement.



22. (a) Option (a) is the best description.
23. (c) Total number of words formed from letters of the word 5!
 $INDIA = \frac{5!}{2!} = 60$
 60th word is NIIDA
 59th word is NIIAD
 58th word is NIDIA
24. (c) Number of triangles $16+8+4+4+436$
25. (b) India won means Sachin is not the captain.
26. (a) In numbers from 1 to 100, 6 appears 10 times at unit place and also 10 times at ten's place.
 Thus, by replacing 6 by 9, variation in the sum is
 $3 \times 10 + 3 \times 10 \times 10 = 330$
27. (b) Let x be the number of steps in staircase.
 Then, $\frac{x-26}{30} = \frac{x-34}{18}$
 $\Rightarrow \frac{x-26}{5} = \frac{x-34}{3}$
 $\Rightarrow 3x - 78 = 5x - 170$
 $\Rightarrow 2x = 92$
 $\Rightarrow x = 46$
28. (b) Let the required numbers be xy.
 Then, $(yx) - (xy) = 18$
 $10y + x - 10x - y = 18$
 $\Rightarrow 9(y - x) = 18$
 $\Rightarrow y - x = 2$
 Hence, possible numbers are 24, 35, 46, 57, 68, 79 besides 13.
29. (c) 1st, 2nd, 4th, 5th and 6th letters of REASONING RESON: It will be rearranged to form NORSE (NE) is the first and last letter.
30. (b) White hair, long tail and a collar will give one truth and one lie in each of the three statements.
31. (c) Let x km/h be the speed of the train and y km be the distance.
 Then, $\frac{y-60}{\frac{3}{4}x} = \frac{25}{x} + \frac{y-85}{\frac{3}{4}x} + \frac{10}{60}$
 $\Rightarrow \frac{25}{\frac{3}{4}x} = \frac{25}{x} + \frac{1}{6}$
 $\Rightarrow \frac{25}{3x} = \frac{1}{6}$
 $\Rightarrow x = 50 \text{ km/h}$
 and $y = 160 \text{ km}$
32. (c) In ADE ABC, only one candidate is repeated and B in there with A
33. (b) In CDE, exactly one is repeated.
34. (a) Only statement I is correct.
35. (b) Due to D there is a repetition, so rest two will be new members.
36. (d) Only two 5 satisfies the requirement.
37. (b) According to Pie chart, profit is reduced by
 $100000 \times 20\% \times 20\% = \text{Rs.}4000$
38. (a) Required price = $100000 \times 20\% \times 15\% \times 25\% = \text{Rs.}750$
39. (a) Percentage contribution of transmission cost
 $= \frac{20 + 20 \times 10\%}{50 + 20 + 20 \times 10\% + 30 + 30 \times 20\%} \times 100\%$
 $= \frac{22}{108}\% = 20\% \text{ (approx)}$
40. (c) If C was the cost and p was the profit,
 then new cost = $\frac{11C}{10}$
 Profit = $0 - 10$
 Profit is reduced by the amount $\frac{C}{10}$
 Percentage reduction = $\frac{C/10}{C} \times 100\% = 10\% C$
41. (b) Required profit percentage

- $= \frac{C/10}{11C/10} \times 100\%$
 $= \frac{100}{11}\% = 9.09\%$
42. (a) The order of answering questions is as follows 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40, 43, 46, 49
 1st reversal: 50, 47, 44, 41, 38, 35, 32, 29, 26, 23, 20, 17, 14, 11, 8, 5, 2
 2nd reversal: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48
43. (c) From the above last question answer is 48.
44. (d) $2017 + 3$, so 3rd question of first reversal.. 44 is the required 20th question answered by Rajita.
45. (b) In A_{20} , there will be $2 \times 20 - 1 = 39$ numbers in AP with common difference 2.
 Also, the first term of A_{20} will be
 $3 + 2 \times 3 + 2 \times 5 + 2 \times 7 + \dots + 2 \times 37 + 2$
 $= 3 + 2(1 + 3 + 5 + 7 + \dots + 37)$
 $= 3 + 2 \times \frac{19}{2} [2 + 18 \times 2] = 3 + 2 \times 19 \times 19 = 725$
 Average of 39 numbers of A_{20}
 $= \frac{39}{2} [2 \times 725 + (39 - 1) \times 2] = 725 + 38 = 763$
46. (c) According to the statements, Venn-diagram is
Conclusions
 I. $\checkmark$
 II. $\checkmark$
 III. $\times$
47. (d) According to the statements, Venn-diagram is
Conclusions
 I. $\checkmark$
 II. $\times$
 III. $\checkmark$
48. (a) According to the statements, Venn diagram is
Conclusions
 I. $\checkmark$
 II. $\checkmark$
 III. $\checkmark$
 IV. $\checkmark$
49. (a) According to the statements, venn-diagram is
Conclusions
 I. $\times$
 II. $\checkmark$
 III. $\times$



Solutions (Q. Nos. 50-54)

House	A	B	C	D	E
Roof		R			G
Chimney	W	B	W		

50. (b) A and C have white chimney.
51. (C) A and C have white chimneys and E may also have white chimney.
52. (c) A, C and E can have green roofs.
53. (a) All three combinations are possible.
54. (a) House E has a white chimney.
55. (d) As, $3^2 < 13 \leq 3^3$ and further it is not known that ball is lighter or heavier, so, minimum number of weighing required is $3 + 1 = 4$





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NIMCET ORIGINAL
PAPER 2009
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